

Transitioning from illegal rooftop dwellings to solar PV: Market-based incentive design and techno-economic analysis

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ABSTRACT

Renewable energy plays a crucial role in achieving net-zero emissions. As an island with a high population density, Taiwan has increasingly favored rooftop solar photovoltaic (PV) systems over ground-mounted installations due to land-use competition. However, illegal rooftop dwellings in urban areas pose significant obstacles to renewable energy deployment. This study proposes a market-based incentive scheme for rooftop solar adoption to encourage property owners to shift away from illegal rooftop dwellings. We construct market supply curves for illegal rooftop rental housing based on real-world rental data and conduct a techno-economic analysis to examine the required government subsidy. The results provide empirical evidence for an important nonlinearity in the supply curves, implying that linearity-based policy frameworks may not be effective in achieving optimal results. We also identify that opportunity costs linked to rental loss constitute a substantial portion of total costs, deterring the owners from adopting the shift. The findings indicate that the existing feed-in-tariff policy is inadequate to incentivize property owners to deploy rooftop solar panels. The proposed scheme embraces the three dimensions of sustainable development—economic, social, and environmental, which could benefit cities worldwide with illegal rooftop structures.

1. Introduction

Climate change is one of the major challenges of the international community. Carbon emissions must be reduced by 45% by 2030 from 2010 and reach net zero by 2050 to limit the global temperature increase to no more than 1.5 °C, avoiding the worst climate damages [1]. Asia-Pacific, accounting for 45% of global greenhouse gas (GHG) emissions, plays a key role in achieving net zero [2]. The large carbon emitters, China, Japan, and South Korea, have all committed to net-zero emissions by 2060, 2050, and 2050 [3–5]. Taiwan, although not a signatory to the Paris Agreement, has also followed its Asian counterparts and proposed “Taiwan’s Pathway to Net-Zero Emissions in 2050 (the 2050 Net-Zero pathway)” in 2022 to reach net-zero emissions by 2050 [6].

The success of the global net-zero transition depends on the energy system transformation from fossil-based to sustainable sources—one of 17 Goals that make up the 2030 Agenda for Sustainable Development [7]. With the global efforts on sustainable energy transition, renewables had expanded and accounted for 29% of global electricity generation by 2020 [8]. As for Taiwan, however, the share of renewables in electricity

generation only reached 6% in 2021 [9]. The Taiwanese government has set ambitious goals for near-term solar energy adoption to achieve the climate targets—20 GW of solar photovoltaic (PV) installed capacity by 2025 (nearly 2.6 times the 2021 level), followed by an annual increase of 2 GW until 2030 [6]. Initially, the 20 GW solar target was set to be 3 GW rooftop and 17 GW ground-mounted systems [10,11]. However, the deployment of ground-mounted solar PV systems has been impeded by land-use competition, driving the government to raise the capacity target of the rooftop type to 8 GW in 2022 while reducing the ground-mounted type to 12 GW [12].

Nevertheless, one of the major challenges when deploying rooftop solar PV systems in Taiwan is illegal rooftop dwellings. Illegal structures have been widely built and occupy the top of residential buildings in Taiwan, particularly in metropolitan areas. Among a total of 711,109 cases of illegal construction, nearly 84% (595,478 cases) are located in the six special municipalities (i.e., Taipei City, New Taipei City, Taoyuan City, Taichung City, Tainan City, and Kaohsiung City) due to the scarcity of space [13]. These unauthorized structures occupy valuable rooftop spaces that could otherwise be utilized for legally deploying solar panels to generate clean energy. The encroachment of these illegal

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constructions obstructs the widespread implementation of rooftop solar PV systems, inhibiting their potential contribution to the renewable energy landscape. Addressing the issue of illegal rooftop dwellings becomes crucial not only from a regulatory standpoint but also in unlocking the full capacity of urban rooftops for solar PV deployment. The illegal rooftop dwellings history can be traced back to 1949 when the Chinese Nationalist Party (i.e., Kuomintang) retreated from mainland China to Taiwan [14]. Urban areas witnessed a massive influx of migrants, leading to severe housing shortages [15], and the rapid population expansion further exacerbated the problem of insufficient living space. Consequently, illegal rooftop dwellings emerged as a solution for the growing housing needs. Facing double jeopardy—external military threats and internal subversion, the Taiwanese government took a passive attitude towards the evolution of illegal rooftop constructions until the 1990s [15,16].

Over the past decades, the governments have cracked down on illegal rooftop dwellings more and more proactively for public safety. However, such problems still exist nowadays and are hard to eradicate in Taiwan. The owners consider those illegal properties their private assets, making it challenging for the government to enforce the law and tear down the violations despite their illegality. Take Taipei City (i.e., the capital city) as an example. The current regulations exempt illegal units on the top of buildings built before 1994 from demolition unless posing obvious hazards to safety or hinder urban renewal implementations [17,18]; similar situations also occur in many other cities [19]. These exemptions have led to the widespread presence of illegally constructed rooftop units throughout urban areas, limiting rooftop availability and potentially impeding the deployment of rooftop solar PV systems. It is worth noting that the issue of illegal rooftop dwellings is not unique to Taiwan. Similar problems are experienced in other regions, such as Hong Kong and Seoul, where developments of illegal rooftop dwellings have arisen due to comparable patterns of economic development and urbanization [20]. In these cities, the governments are aware of the issue but also tolerate the existence of legal gray areas. The presence of conflicting interests among various groups has posed challenges for governments in effectively enforcing law enforcement measures [21]. These complexities surrounding illegal rooftop dwellings in different regions underscore the need for innovative and comprehensive approaches to address the issue and promote sustainable urban development.

Besides rooftop availability, cost is also a major driver of solar PV adoption [22–25]. Burt and Dargusch (2015) [22] examined the economic factors on solar PV uptake in Australia, suggesting that the most critical determinants are the PV panel costs, feed-in tariffs (FIT), and renewable energy credits. Mah et al. (2018) [23] investigated residents' perceived barriers to solar PV deployment in Hong Kong—one of the countries with the world's most expensive property market and the most severe problem of illegal rooftop units. The study identified the top three barriers: long payback periods, high upfront costs, and limited rooftop space. And the proposed barrier mitigation strategies include designing a subsidy policy to reduce the upfront costs and payback periods, incentivizing homeowners to free up their occupied roofs for solar PV deployment, and providing adequate technical support for owners. Lee et al. [24] underlined the importance of incentive policies as the initial investment costs for solar PV systems remained high and represented a substantial proportion of the overall costs. The study culminated in the identification of several approaches to enhance state-level solar incentives in the United States. Wen et al. [25] conducted a systematic review of the incentive policies pertaining to China, Japan, Germany, and the United States regarding their promotion of solar markets. Despite varying policies across countries, the investigation revealed a consistent finding that investment subsidies have a positive impact on the expeditious expansion of solar adoption. Most of the existing studies focus on identifying and assessing factors influencing household solar PV adoption [22,23,26,27], sharing the same conclusion that government plays a key role in the clean energy transition—providing subsidies

and actively demonstrating commitments to renewables. Among those studies regarding the development and impacts of solar subsidies and incentives [24,25,28,29], none consider the transition costs from illegal additions to rooftop solar PV systems in the evaluation.

In Taiwan, the government launched the “Green Energy Roofs Project (the Project)” in October 2017, with a principle of “zero expense from citizens and zero allowance from governments” [30]. In the Project, local governments merely serve as coordinators, matching solar PV companies with house owners willing to install solar panels on their roofs [31]. The solar PV companies bear the installation and operational costs while the house owners provide their roof space for PV installation [31]. The house owners can take advantage of solar power to save money on energy bills first. The solar PV companies will then sell the excess power from rooftop solar to the power company at the given solar feed-in-tariffs (FIT) rates [31]. The level of FIT is mainly determined based on the levelized cost of electricity generated from rooftop solar PV [32], ensuring that solar adopters can at least recover all the different PV-related costs. However, under this FIT mechanism design, illegal rooftop unit owners do not have sufficient incentives to tear down those additions for solar deployment. During the transition from illegal rooftop dwellings to solar PV systems, the unit owners would incur not only solar panel-related costs but also other costs—unit demolition and opportunity costs. Demolition of illegal rooftop dwellings provides positive externalities to society, which could justify receiving grants and subsidies from the government. And since the owners of illegal additions usually rent the properties out to earn money, if the units are torn down for solar PV deployment, owners will lose the rental income—so-called opportunity costs.

The existing FIT scheme is, apparently, unable to cover the income gaps faced by the owners of illegal rooftop dwellings, keeping them from removing the additions and adopting solar PV systems. The lack of adequate financial support is constraining the growth of solar power deployment and thus threatening the net-zero target in Taiwan. Therefore, redesigning clean energy subsidies for shifting from illegal rooftop dwellings to solar PV systems becomes a pressing need. Here, we propose a novel incentive scheme to promote renewable energy—particularly for cities with the problem of illegal rooftop structures. Moreover, we construct supply curves of illegal rooftop housing based on real-world rental data and document empirical evidence for market nonlinearity: supply curves are convex. By selecting Taiwan's four largest special municipalities as a case study, we investigate the government subsidy required to close the gaps between the proposed rooftop solar PV systems and the existing illegal rental housing. We then conduct a sensitivity analysis to understand how the outcomes change if major parameters (such as FIT and monthly rental income) are changed. We conclude by stating some implications for policymakers seeking to facilitate the clean energy transition.

2. Methods and data

2.1. Incentive scheme overview

Fig. 1 illustrates the relationship between the stakeholders involved in the incentive scheme. The scheme aims to promote renewable energy use while ameliorating the problems of illegal rooftop dwellings in urban areas. The main idea behind the scheme is that demolition companies should first tear down the illegal rooftop dwellings, and the released rooftop space can then be used for solar panel installation. Once solar panels are installed on the rooftops, the owners start to make money by selling the solar power back to the grid at the feed-in tariff (FIT) rates. Illegal rooftop dwellings usually provide rental income for property owners, so the owners will have rental losses as their illegal additions are removed—such opportunity costs resulting from switching to solar power generation are considered in the scheme. Governments play a crucial role in carrying out the scheme by covering the income gaps faced by the owners of illegal rooftop dwellings.

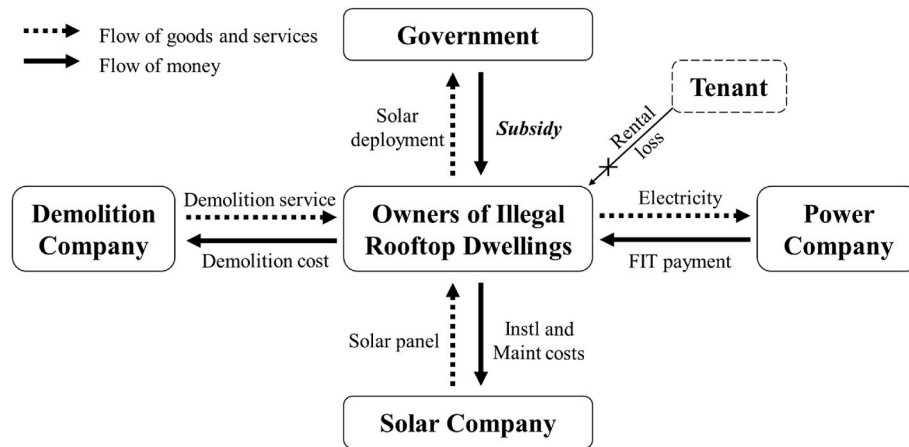


Fig. 1. Relationship diagram between the stakeholders involved in the market-based incentive scheme for replacing illegal additions with solar panel installations.

2.2. Market-based incentive calculation model

In this subsection, we initiate by delineating the cost and benefit components linked to the replacement of illegal rooftop dwellings with solar panels, as borne by the property owners. Subsequently, we introduce the solar deployment incentive evaluation model, designed to equilibrate the costs and benefits for the owners of these illegal additions. Finally, we provide a concise summary of all the critical parameters and their respective data sources essential for computing the required government subsidy. For brevity, we focus only on the four largest special municipalities with the population greater than 2.5 million in Taiwan—Taipei City (Northern Taiwan), New Taipei City (Northern Taiwan), Taichung City (Central Taiwan), and Kaohsiung City (Southern Taiwan). Fig. 2 summarizes the demographic and socioeconomic information for these selected cities. However, the method we illustrate below makes it straightforward to study the market-based incentives for rooftop renewable energy deployment in other regions.

2.2.1. Cost and benefit components

Switching from illegal rooftop dwellings to solar power generation requires an upfront investment in solar panel equipment and installation and the demolition costs of illegal rooftop constructions. After that, the only recurring charge is the solar PV system maintenance. Besides these explicit costs, we also consider the implicit cost – opportunity cost – arising from choosing solar energy deployment over extra rental space. On the other hand, the owners could benefit from having solar panels on their roofs through the feed-in-tariff (FIT) policy mechanism. Each of the cost and benefit components is described below.

- (1) **Demolition cost** is the cost for demolition companies to tear down illegal rooftop dwellings. The Taipei City and New Taipei City governments have established fee-charging standards for the demolition of unauthorized structures [35,36]. We select the most common construction type with brick materials as the base case, with the corresponding demolition cost of 620 NTD/m². The demolition costs vary depending on the material and structure of the construction, ranging from 200 NTD/m², the cost of mortar-cobblestone-made construction, to 1,720 NTD/m², the cost of steel construction.
- (2) **Solar panel capital cost** is the upfront cost to purchase solar panels and have them installed on the roofs. It is important to note that larger solar panel systems generally result in higher total costs but lower per-kW costs due to economies of scale. In the case of Taiwan, the rooftop space typically released from the removal of illegal additions is limited to less than 200 m² [37]. Considering the required area for a solar panel system, approximately 10 m² is necessary to accommodate a 1 kW capacity of the solar PV system

[38]. With this information, we proceed to calculate the solar panel costs utilizing the per-unit cost based on the government's FIT scheme, which provides a rate of 55,600 NTD/kW for installed capacities of less than 20 kW [39].

- (3) **Solar panel maintenance cost** is the cost paid regularly to ensure the smooth operation of solar panels. With the released rooftop space smaller than 200 m² and the total installed capacity less than 20 kW, the suggested maintenance cost is 2,454 NTD/kW based on the FIT scheme [39].
- (4) **Opportunity cost** is the rental income loss by the owner owing to the removal of illegal additions. We gather the renting information of illegal rooftop dwellings from the local leading apartment matching platform, and a total of 13,023¹ rental property rates are acquired for the analysis [37]. The distribution of the collected rental rates is utilized to derive a market supply curve for illegal rooftop properties, as shown in the Result and Discussion Section. The medium rates serve as the basis for opportunity cost calculations—368.0, 257.1, 221.0, and 211.8 NTD/m² for Taipei City, New Taipei City, Taichung City, and Kaohsiung City, respectively.
- (5) **Rooftop solar energy benefit** is FIT payments owners can get when they sell solar energy back to the grid. Annual FIT income is calculated by multiplying the FIT rate by the annual electricity generated by solar panels and fed into the grid power. Based on the solar panel capacity size and category, the FIT rates vary from 3.8680 to 5.8952 NTD/kWh. This study uses the FIT rate of 5.8952 NTD/kWh set by the government for the installed capacity of less than 20 kW. A solar panel's electricity output depends on many factors including not only capacity size, but also installed location and weather conditions. On average, the annual electricity generation from a 1 kW solar panel system is higher in Southern Taiwan than that in Northern Taiwan: 1,002, 989, 1,276, and 1,162 kWh/kW for Taipei City, New Taipei City, Taichung City, and Kaohsiung City, respectively [40]. Due to lower solar electricity adoption, the government offers more incentives for residents in northern Taiwan (including Taipei City and New Taipei City)—who are eligible to sell solar power at a premium of up to 15% compared with the national FIT rate [32].

2.2.2. Government incentive calculation

Government incentives are strategically devised to bridge the gap between rental income derived from illegal additions and the net ex-

¹ This collected dataset comprises 13,023 illegal rooftop rental properties across four cities, representing approximately 30% of the total rental properties available in these areas.

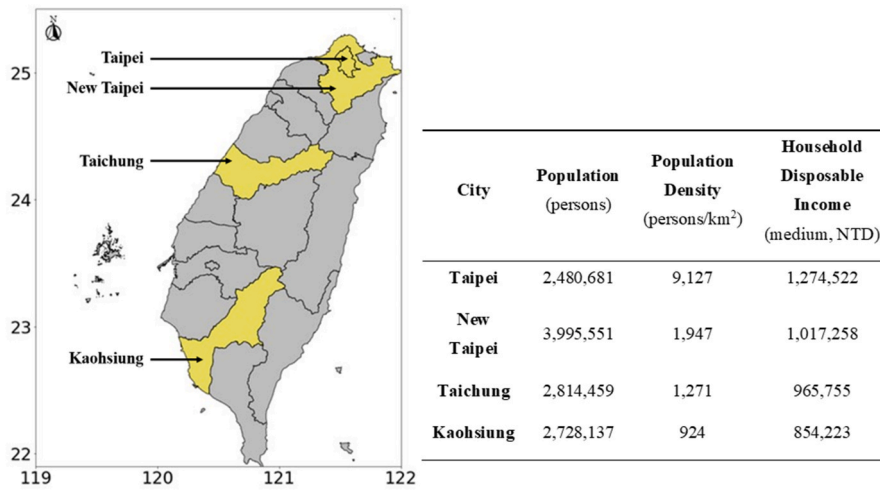


Fig. 2. Summary of the demographic and socioeconomic information for the selected cities [33,34].

penses associated with solar deployment. To calculate the subsidy aimed at encouraging the transition from illegal rooftop structures to renewable energy generation, we utilize Equation (1), wherein the loss of rental income is treated as a cost, representing the opportunity cost, and the FIT payment for solar energy generation is regarded as an owner's benefit resulting from the proposed transformation.

$$\text{Subsidy} = \text{Costs} - \text{Benefits}$$

$$= \left(A c_d + c_s + \sum_{i=1}^N \frac{c_m}{(1+r)^i} \right) + A \times \sum_{i=1}^N \frac{12c_o}{(1+r)^i} - \left(\sum_{i=1}^N \frac{\text{FIT} \times G \times (1 - \text{DR} \times (i - 10))}{(1+r)^i} \right) \quad (1)$$

where A is the required area for a 1 kW solar panel system (m²/kW); c_d is the demolition cost (NTD/m²); c_s is the unit solar panel capital cost (NTD/kW); c_m is the annual solar panel maintenance cost (NTD/kW-year); c_o is the opportunity cost for monthly average rental loss (NTD/m²-month); r is the discount rate (%); N is the lifespan of a solar panel system (year); FIT is the feed-in-tariff rate (NTD/kWh); G is the annual solar power generation, depending on the deployment location (kWh/kW); DR is the degradation rate (%), which is assumed to be 0 within 10 years and 1% per year after that (i.e., $\text{DR} = 0$ if $i \leq 10$; = 1% if $i > 10$) [39].

2.2.3. Data and governing parameters

In this subsection, we provide a detailed account of the data sources and governing parameters used in our incentive evaluation for rooftop solar PV deployment. The parameters associated with solar panels were derived from the FIT calculation scheme established by the government to ensure consistency in our calculations [39]. By utilizing these official parameters, we can maintain accuracy and align our study with the government's renewable energy policies. Table 1 serves as a comprehensive compilation of all the crucial parameters employed in our analysis, which plays a fundamental role in determining the feasibility and effectiveness of the proposed incentive scheme.

3. Results and discussion

3.1. Supply curves of illegal rooftop dwelling rental market

Understanding the relationship between the rental rates and the quantities of rooftop housing illegally supplied on the market is crucial to designing an incentive scheme for rooftop solar deployment. To construct the supply curves of the illegal rooftop dwelling markets, we

Table 1

Governing parameters applied in the government incentive evaluation and the sources.

Parameter	Value	Source
Costs		
Demolition Cost (NTD/m ²)	620	[35,36]
Unit Solar Panel Capital Cost (NTD/kW)	55,600	[39]
Annual Solar Panel Maintenance Cost (NTD/kW·year)	2,454	[39]
Benefits		
Annual Solar Power Generation (kWh/kW)		
Taipei City	1,002	[40]
New Taipei City	989	
Taichung City	1,276	
Kaohsiung City	1,162	
Feed-in-tariff Rate (NTD/kWh)		
Taipei City and New Taipei City	6.77948	[32]
Taichung City and Kaohsiung City	5.8952	
Area per kW solar panel system (m ² /kW)	10	[38]
Lifespan of Solar Panels (year)	20	[41]
Degradation Rate (%)		
Within 10 years	0	[39]
After 10 years	1	
Other Parameters		
Discount Rate (%)	5.25	[39]

adopt the following steps: Firstly, each illegal rental unit is represented by a horizontal line, its length proportional to the unit's area. Next, these lines are arranged on a plot from left to right, without overlapping, based on the monthly rents per square meter of the units, sorted from lowest to highest. Then, the vertical coordinate of each line is adjusted according to the respective monthly rents per square meter. Finally, we connect neighboring horizontal lines with vertical lines, resulting in the derivation of the supply curves for the illegal rooftop dwelling markets in the four selected cities, as illustrated in Fig. 3.

The results suggest that the market supply of illegal rooftop properties differs significantly between northern Taiwan and central and southern Taiwan. The total (cumulative) quantities of illegal rooftop dwellings in Taipei City and New Taipei City are much greater than in Taichung City and Kaohsiung City. As parts of the largest metropolitan area in Taiwan, Taipei City and New Taipei City provide the most high-quality education and job opportunities, thus attracting a large number of migrants from other cities. In 2020, immigration accounted for 6.6% and 8.5% of the population in Taipei City and New Taipei City, respectively [42]. Given the limited space in the urban cities and high property prices, population influx has driven more homeowners to build rooftop additions on their apartments to create more space. These rooftop additions are generally unauthorized or illegally constructed but

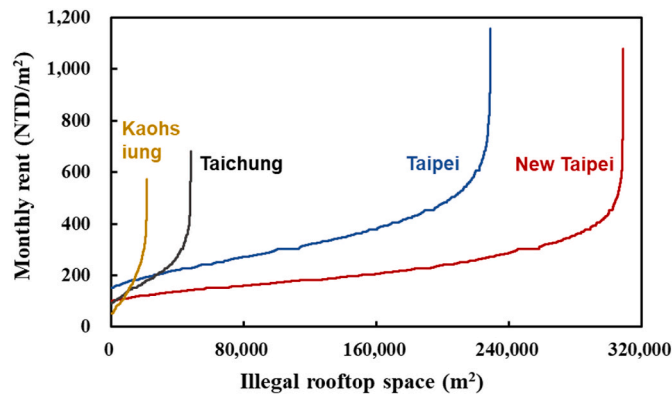


Fig. 3. Supply curves of illegal rooftop dwelling rental market in Taipei, New Taipei, Taichung, and Kaohsiung cities in Taiwan.

more affordable than others with approval from building regulations. Based on the collected dataset, the average monthly rents of illegal rooftop dwellings in Taipei City, New Taipei City, Taichung City, and Kaohsiung City are 386.4, 271.8, 229.5, and 212.4 NTD/m²—which are about 4.1%–17.5% lower than those of the whole rental markets at the city level, 468.2, 303.2, 240.5, and 221.6 NTD/m², respectively [43].

Furthermore, Fig. 3 provides empirical evidence for nonlinearity in illegal rooftop dwelling rental markets—supply curves are flat at low levels and nearly vertical at very high levels of capacity utilization. In other words, the supply curves of the illegal rooftop dwelling rentals are convex. Capacity constraints at the property level result in the steepness of the supply curve close to the market capacity. Although the quantities supplied vary a lot by city, there are no significant differences in the shapes of supply curves of illegal rooftop housing—which all slope up and are convex. The findings of the nonlinear supply curves of illegal rooftop dwelling markets will lead to several implications for the following incentive scheme design discussion. Readers should note that illegal rooftop housing supply curves do not start from the origin because they are not unit elastic. Renting out properties takes time and effort, sometimes considered more of a headache than it is worth. Thus, property owners would leave the illegal additions idle unless rental prices exceed certain thresholds—approximately 150, 100, 90, and 50 NTD/m² for Taipei City, New Taipei City, Taichung City, and Kaohsiung City, respectively.

3.2. Market-based subsidy scheme for shift from illegal rooftop dwellings to solar panels

Based on the data presented in Fig. 3, we establish the relationship between the required government subsidy and rooftop solar power

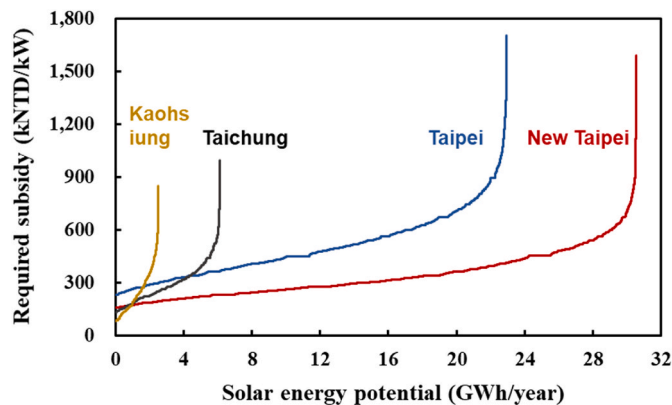


Fig. 4. Required government subsidy amount as a function of annual rooftop solar energy in Taipei, New Taipei, Taichung, and Kaohsiung cities in Taiwan.

potentials, as depicted in Fig. 4, through the following conversion steps. For the independent variable (x-axis), the annual solar power generation is determined by multiplying the areas of illegal rooftop properties (x-axis in Fig. 3) by the annual solar power generation per unit area of the solar panel system (G/A in Equation (1)), where G varies based on the deployment location. On the other hand, for the dependent variable (y-axis), the required government subsidies, dependent on the rental loss (co ; NTD/m²·month), are calculated using Equation (1). It is important to note that all illegal rooftop spaces to be released will be dedicated to solar panel deployment in our proposed scheme.

Fig. 4 presents the relationship between the required government subsidy and annual rooftop solar power generation additions, reflecting the nature of the illegal rooftop dwelling markets—nonlinearity. Capacity constraints at the property levels generate convex supply curves at the market levels, causing the required government subsidy to increase sharply when close to capacity. For example, fewer illegal rooftop structures are for rent in Taichung City and Kaohsiung City, causing their marginal curves of required government subsidies to steep much sooner than in the Taipei metropolitan area (i.e., Taipei City, New Taipei City). While linear approximations are commonly employed for analytical convenience, the linearity assumption can lead to policy failure for illegal rooftop dwelling markets that are fundamentally nonlinear (Fig. 3). For example, linearized models cannot address the policy-relevant question of whether the impacts of a fiscal policy vary with the changing market demand. A linear framework implies that a subsidy scheme would have the same effectiveness in encouraging a shift to rooftop solar PV systems regardless of market expansion or contraction—which is not valid owing to the convexity of the supply curves of illegal rooftop apartments.

Table 2 displays the city-level comparisons between the solar rooftop potential identified in this study and the cumulative installed solar capacity at the end of 2021. The key observations are highlighted as follows. Firstly, rooftop solar potentials that the proposed subsidy scheme could realize are much greater in the Taipei metropolitan area, owing to its ubiquitous illegal rooftop housing. Secondly, solar deployment is geographically diverse. With 773 MW, Kaohsiung City has the most installed solar capacity, followed by Taichung City. However, the solar installations in Taipei and New Taipei are much fewer than in Taichung and Kaohsiung. Thirdly, illegal rooftops could significantly boost urban solar power by 57% in Taipei City and 44% in New Taipei City. In contrast, replacing all the illegal rooftop housing with solar PV systems could only add 1.1% and 0.3% in Taichung City and Kaohsiung City, respectively.

3.3. Government subsidy breakdown

To understand the composition of the government incentives, we break down the income and expenses incurred by the illegal housing owners who opt to deploy solar panel systems. For illustrative purposes, we use the medium rental rates as the basis for opportunity cost calculations. Fig. 5 compares the sum of the present values of all costs and

Table 2

City-level comparisons of rooftop solar capacity potential derived from illegal additions and the cumulative solar installations at the end of 2021.

City	Illegal Rooftop Housing [A] (m ²)	Rooftop Solar Capacity Potential ^a [B] = [A]/10 (kW)	Installed Solar Capacity ^b (kW)
Taipei	229,021	22,902	40,515
New Taipei	308,821	30,882	70,907
Taichung	47,968	4,797	437,190
Kaohsiung	21,600	2,160	773,271

^a The required area per kW solar panel system is 10 m²/kW [38].

^b We show the combined rooftop and utility-scale solar capacity due to the lack of breakdown data at the city level.

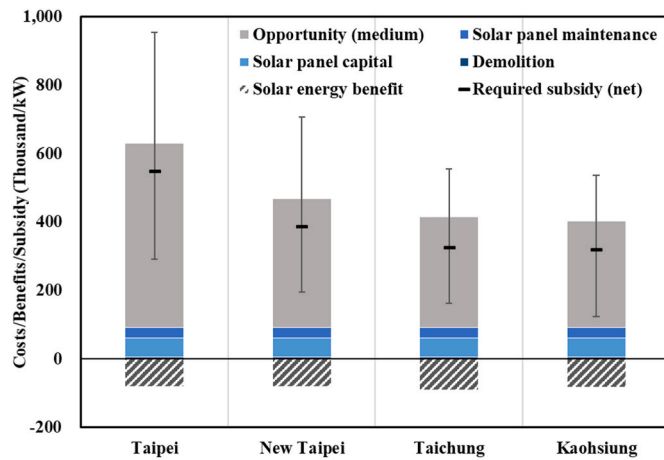


Fig. 5. Government subsidy breakdown per kW of rooftop solar PV deployment for the proposed incentive scheme by the city; error bars are from differences in monthly rental income of illegal rooftop dwellings.

benefits on a per-kW basis across four selected cities, revealing several findings as highlighted below. Notably, our analysis focuses on major expenditures involved in the transformation processes of illegal rooftop structures, rather than attempting to encompass all cost and benefit components. Consequently, certain minor expenditures have been excluded from our calculation. For instance, moving costs, which are typically borne by tenants, may, in some cases, be covered by landlords when providing furniture to tenants during move-in. Additionally, we have excluded the cost of finishing paperwork for the application of government subsidies. It is crucial to emphasize that the exclusion of these minor expenditures does not introduce significant bias, and as such, they are not factored into our final calculations.

Firstly, opportunity costs associated with rental losses are the most significant contributors, accounting for 77–85% of the total per-kW costs depending on the cities; the contribution is higher in the Taipei metropolitan area owing to the higher rental prices. This observation highlights the importance of accounting opportunity costs in renewable subsidy designs; otherwise, the deriving incentives are likely not adequate for rooftop owners to cover the income gaps between rental properties and solar power. Secondly, solar panel capital costs are the second largest component, followed by maintenance costs. Demolition costs are fairly insignificant—about an order of magnitude smaller than solar panel-related costs and two orders smaller than opportunity costs. Thirdly, besides opportunity costs, all the other cost components are assumed to be the same across all cities in this study due to data availability. Although demolition costs for illegal rooftop structures could vary a lot under different circumstances (e.g., scale, complexity, demolition permits, and labor costs), this assumption would not introduce substantial bias in the calculations due to their minor contribution. Fourthly, Taichung City could benefit most on a per-kW basis thanks to the greatest (10–29% higher) kWh of solar energy generated from a kW of solar power in Taichung compared to the other cities. Despite having a 15% premium on the FIT rate, per-kW rooftop solar energy benefits in the Taipei metropolitan area are still about 11% less than in Taichung City. Finally, subsidies are derived using Equation (1), identifying the differences between costs and benefits (i.e., net costs). The results show that Taipei City requires the greatest per-kW incentives (i.e., 549,343 NTD/kW) to motivate the shift from illegal rooftop dwellings to rooftop solar panel systems—42%, 69%, and 72% more than New Taipei, Taichung, and Kaohsiung, respectively.

In addition to our research findings, we acknowledge that among the four cities studied, Taipei is the only city with a long-term policy on the removal of old illegal dwellings constructed before 1963 [44]. Under this policy, owners are eligible for subsidies at a rate of 5,000 NTD/m² for the removal of illegal additions. Moreover, owners may also receive a

relocation fee of 45,000 NTD per household. For those who proactively remove their illegal additions, an additional 100,000 NTD per household may be granted. Considering the data collected, the average area of illegal rooftop dwellings in Taipei is approximately 40 m². Hence, if the aforementioned conditions are all met, an owner may receive a total subsidy of 8,625 NTD/m² for removing their illegal additions. In comparison, the rental loss in Taipei for the same area is approximately 53,885 NTD/m². This comparison provides valuable context on the disparity between the existing policy measures in Taipei and the actual rental losses faced by property owners due to the removal of illegal additions. It underscores the potential financial burden property owners may encounter when attempting to comply with the city's policy. By highlighting this insufficiency in the existing policy, our study emphasizes the importance of additional financial incentives, such as the proposed market-based incentive scheme, to bridge the gap and encourage property owners to transition from illegal rooftop structures to rooftop solar PV systems.

3.4. Climate change mitigation benefits

To understand the aggregate effects of the proposed government incentives on carbon reduction, we estimate the total energy outputs that could be produced from the derived rooftop solar PV systems instead of the existing power grid. Using the location-specific per-kW solar energy output factors, we calculate the annual rooftop solar power generation potential at the city level, as displayed in Table 3 [9]. In total, the transition from identified illegal additions to rooftop solar panels could lead to an additional generation of approximately 62.12 GWh of solar energy per year. While this amount of solar energy represents only 0.23% of residential electricity consumption in these cities for the year 2021, it holds significant potential to advance clean and sustainable energy generation in the region, accounting for an impressive 4.74% of the existing solar PV generation in the four cities [9]. Assuming the electricity carbon intensity is fixed at 0.509 kgCO₂/kWh [45], this transformation will cause the cumulative carbon emissions from household electricity consumption to decrease by 0.63 Mt over the solar panel lifespan of 20 years. We monetize the benefits of these carbon emission reductions using the social cost of carbon dioxide of 185 USD/tCO₂ [46] and find that a total of 3,591.4 million NTD savings could be accumulated and realized for the four selected cities. At the city level, the contributions to global climate change mitigation are 1,326.8 million NTD for Taipei, 1,765.9 million NTD for New Taipei, 353.9 million NTD for Taichung, and 144.8 million NTD for Kaohsiung, respectively.

On the other hand, the climate change mitigation effects come at a cost to the government. By multiplying the potential rooftop solar capacity by the medium estimates of per-kW required subsidy (Fig. 4), we obtain the expenditures for the government to stimulate the transition by city—12,581 million NTD for Taipei, 11,982 million NTD for New Taipei, 1,561 million NTD for Taichung, and 690 million NTD for Kaohsiung. Combining the quantified climate change mitigation benefits with the costs, we assess the cost-effectiveness of the proposed

Table 3

City-level annual rooftop solar power generation potential derived from illegal additions.

City	Rooftop Solar Capacity Potential [B] (kW)	Annual Solar Power Generation Factor [C] (kWh/kW)	Annual Rooftop Solar Energy Potential [D] = [B] × [C] (kWh/yr)
Taipei	22,902	1,002	22,947,804
New Taipei	30,882	989	30,542,298
Taichung	4,797	1,276	6,120,972
Kaohsiung	2,156	1,162	2,505,272
Sum	60,737	–	62,116,346

incentive scheme at the city level. The resulting benefit-to-cost ratios range from 0.1055 in Taipei and 0.1474 in New Taipei to 0.2099 in Kaohsiung and 0.2267 in Taichung. Although the benefits of reduced carbon emissions are less than the expected costs incurred by the governments, we do not quantify all of the benefits of the proposed incentive scheme. For example, removing illegal rooftop dwellings would enhance public safety, and renewable energy expansion would improve national energy security and air quality. Moreover, it should be noted that these monetized carbon reduction benefits have significant uncertainties. There is enormous uncertainty in evaluating the true costs of damages to society of having one extra tonne of carbon emissions. For example, IPCC suggests carbon prices to be 135–6,050 USD/tCO₂ for the 1.5 °C limit goal (IPCC, 2018, p.152). Albeit with substantial uncertainties, this study points out that the social benefits are likely to be comparable to the government expenditures, hoping to foster more market-based incentive scheme studies for renewables elsewhere in the coming decades.

3.5. Sensitivity analysis

The subsidy evaluations presented above rely on several variables (Table 1)—including discount rate, solar feed-in tariff (FIT) rate, structure demolition costs, capital costs and recurring maintenance costs of solar panels, and monthly rental income for illegal rooftop dwellings. We conduct sensitivity analysis to assess how much the required subsidy outputs are conditioned by the latter. For brevity, we focus only on the sensitivity of the medium estimates to assumptions; however, the method described here makes it straightforward to analyze the sensitivity of the spread of the subsidy outcomes.

Fig. 6 presents the change in government subsidy as the major parameters change. As expected, all four cities show similar trends in terms of relationships between government subsidy required and other

variables. We observe that the government subsidy required positively correlates to all the cost terms, including opportunity costs (i.e., monthly rents)—which is not surprising because greater subsidy amounts are needed to compensate for rising costs. For example, as rental rates increase, the opportunity costs (i.e., rental income loss) resulting from the removal of illegal addition increase, thereby the government financial support should be expanded to cover wider gaps between rental income and the net costs of solar deployment. By contrast, the renewable FIT and discount rates have negative relationships with the required subsidy. As the FIT rate increases, the rooftop solar energy benefits increase because owners can earn more money when selling solar energy back to the grid. In this case, less government subsidy is required to compensate for the transition costs. The discount rate, used to calculate the present values of costs and benefits, dictates the relative importance of the upfront capital investment versus annual earnings and recurring expenses. Increasing the discount rate would greatly reduce the value of the future rental income loss and thus lower the required government subsidy. Among all the investigated variables, the required subsidy is most sensitive to the monthly rents of illegal rooftop dwellings—the largest cost component (Fig. 5), followed by the discount rate while being least sensitive to demolition costs, which contributes marginally to the total transition costs.

4. Conclusion

This study addresses the urgent need to propose a market-based incentive scheme, aiming to steer rooftop usage away from illegal additions and towards solar PV implementation to achieve net-zero targets. To achieve this, we collect real-world data on the illegal rooftop dwelling rental market in Taiwan's four largest special municipalities and constructed supply curves. The analysis reveals that the market size is significantly larger in northern cities, primarily due to factors such as

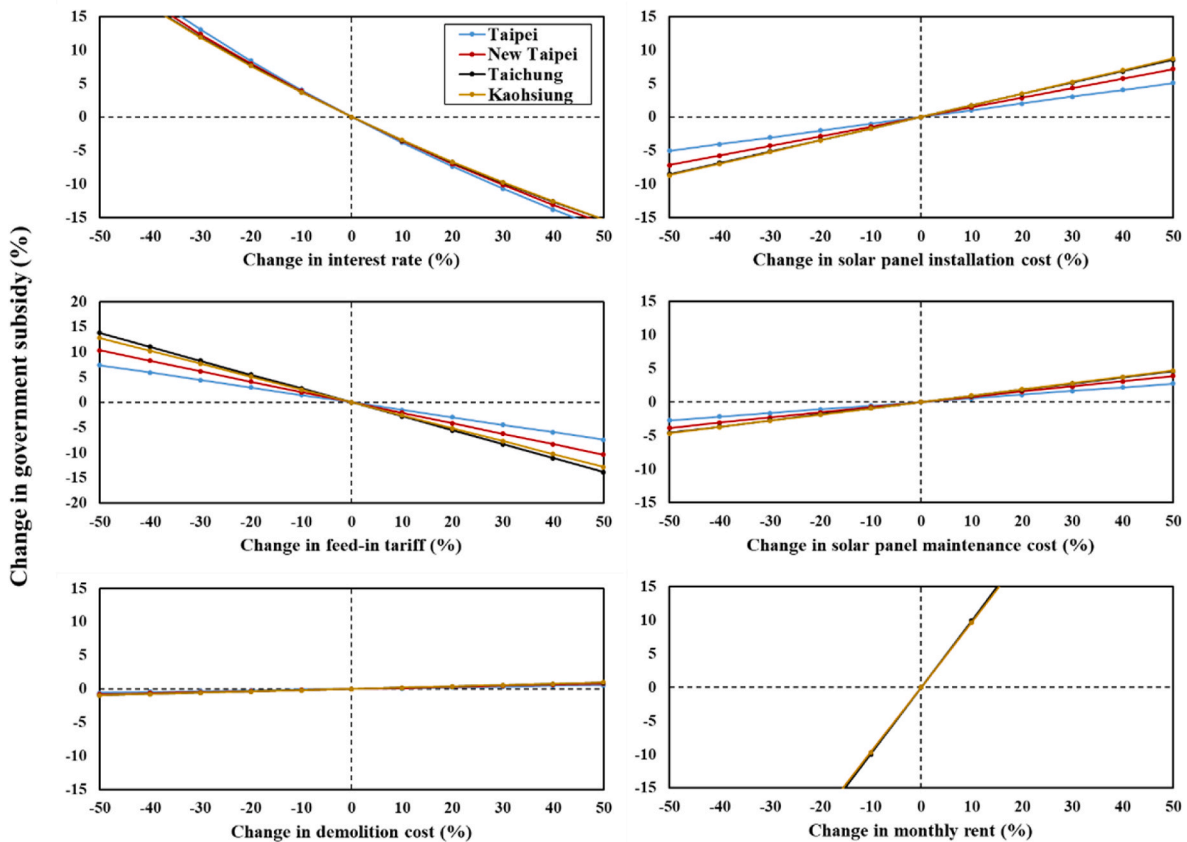


Fig. 6. Sensitivity of required government subsidy with respect to major parameters.

population influx, limited spatial availability, and a strong demand for affordable housing in Taipei and New Taipei compared to Taichung and Kaohsiung. Additionally, the supply curves of illegal rooftop dwelling rentals exhibit a convex and nonlinear nature, indicating steep slopes near market capacity, indicative of capacity constraints in these markets. These findings provide valuable insights into the market dynamics of illegal rooftop dwellings and underscore the necessity of formulating appropriate policies to address capacity constraints in the illegal rental space.

We then establish relationships between the required government subsidies and annual rooftop solar power generation through numerical transformations of the illegal rental market supply curves. The resulting relationships exhibit similar nonlinearity characteristics as the supply curves, highlighting the importance of developing policies that consider the diminishing marginal benefits of financial incentives offered to property owners. As the market capacity is approached, the marginal benefit of subsidies decreases, making their effectiveness less significant. Moreover, a linear policy framework assumes that subsidies have an equal impact on promoting transitions to rooftop solar PV systems, regardless of market expansion or contraction. However, given the convexity of the market, such an assumption becomes invalid, rendering linear-based policies suboptimal and leading to economic inefficiencies. Thus, adopting a market-based approach that accounts for the nonlinear nature of the market is crucial in developing efficient policies to promote the transition effectively.

Furthermore, we conduct an analysis of the costs and benefits incurred by illegal housing owners choosing to install rooftop solar PV to understand the composition of necessary government incentives. Presently, the Taiwanese government incentivizes solar PV adoption through feed-in tariff (FIT) rates that differ between northern and southern regions due to varying insolation levels. However, despite the higher FIT rate in northern Taiwan, solar PV deployment in Taipei and New Taipei lags significantly behind that of Taichung and Kaohsiung. Upon incorporating opportunity costs of rooftop space into our scheme, it becomes evident that these costs represent a substantial share of the overall transition costs, significantly influencing the decision-making of property owners. The existing FIT policy is designed to recoup the costs associated with solar PV adoption, but it proves inadequate in motivating property owners, especially in Taipei metropolitan regions characterized by considerably higher rents (and, consequently, opportunity costs) than Taichung and Kaohsiung. These results underscore the necessity of accounting for opportunity costs when designing subsidies to enhance policy effectiveness.

In addition to the primary analysis, we also quantify the external carbon benefits that the proposed incentive scheme could offer through the transition from identified illegal additions to rooftop solar PV systems in the four cities. Our findings indicate that cumulative carbon emissions of 0.63 Mt could be effectively reduced over the 20-year lifespan of the solar panels, resulting in savings of 3,591.4 million NTD when using the social cost of carbon at 185 USD/tCO₂ [46]. This assessment underscores the significant positive environmental impact that the incentive scheme could bring about, contributing to both carbon emission reductions and potential cost savings. Beyond the reduced carbon emissions, it is important to note that the proposed incentive scheme offers numerous other benefits that are challenging to quantify and out of the study scope. For instance, the removal of illegal rooftop housing has the potential to enhance public safety, mitigating risks associated with unauthorized and unsafe structures. Additionally, increasing solar power capacity may contribute to national energy security by expediting the transition from conventional power generation to distributed clean power generation. These broader benefits represent valuable contributions of the proposed scheme to fostering sustainable urban development and energy diversification.

This study encompasses all three dimensions of sustainability—the environment, economy, and society—in a comprehensive manner. Firstly, by promoting sustainable energy, the proposed incentive scheme

serves as a mitigation measure to reduce the negative environmental impacts of human activities, particularly in terms of mitigating CO₂ emissions. Secondly, in terms of the economy, the scheme considers all associated costs and ensures that the required government subsidies are allocated efficiently to bridge the income gaps faced by illegal rooftop unit owners. This approach enhances economic efficiency, making it economically viable for property owners to make the transition to solar PV installations. Thirdly, the proposed scheme encourages the shift to solar PV systems from illegal rooftop units through positive incentives rather than coercive measures. By utilizing incentives, the scheme complements the limited capacity of law enforcement measures taken by the government to address illegal rooftop dwellings. Moreover, this approach fosters mutual trust between citizens and the government, promoting a collaborative and cooperative environment. Overall, the proposed incentive scheme offers an innovative and integrated solution to two significant social issues in Taiwan: promoting sustainable energy and addressing illegal rooftop dwellings. Instead of tackling these social issues individually, this study demonstrates the potential for integrated solutions that synchronize government efforts and enhance overall efficacy.

Despite the valuable insights provided by this study, it is crucial to acknowledge certain limitations and areas for future exploration. Firstly, the lack of official documentation regarding the number of individuals residing in illegal rooftop dwellings hampers a precise characterization of the impact on affected communities resulting from the proposed incentive scheme. The dataset collected, consisting of 13,023 illegal rooftop rental properties, represents approximately 30% of total rentals across four cities. However, accurately determining the number of people who may be displaced remains uncertain and warrants further exploration. Secondly, in our power generation estimation, we do not directly consider the effects of inclining angles and shadings on the amount of power generated by rooftop solar PV systems. While we utilize the capacity factor provided by the local power company for the four cities to estimate power generation, we are aware that power generation may differ between districts within these cities. Unfortunately, due to data availability, it is not feasible to estimate the power generated district by district, and future research should consider more detailed data and analysis to account for these variations. Lastly, while our study provides a practical framework for market-based incentives in the context of Taiwan's four largest special municipalities, it is essential to acknowledge that its applicability and effectiveness may vary in different geographical settings. Future research is needed to explore the feasibility of implementing this scheme in other regions and countries and to consider the specific environmental and climate factors that may influence solar PV generation.

Author contributions

Yu-Jui Chang: Formal analysis, Investigation, Writing - Original Draft, Visualization. I-Yun Lisa Hsieh: Conceptualization, Methodology, Supervision, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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